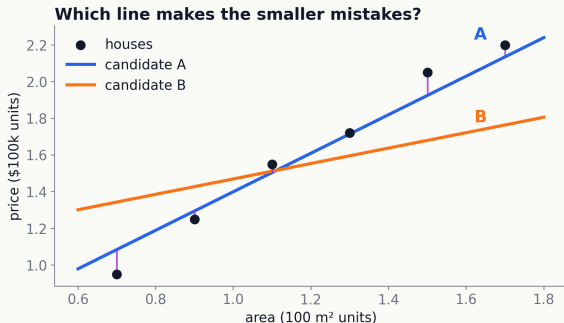


Can a line learn house prices?

A beginner-friendly journey through predictions, mistakes, and updates.

Today we will calculate one complete learning step.



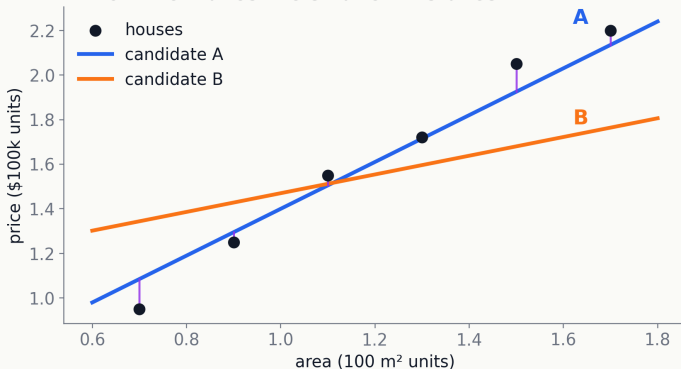
Which line is better?

How could a machine decide?

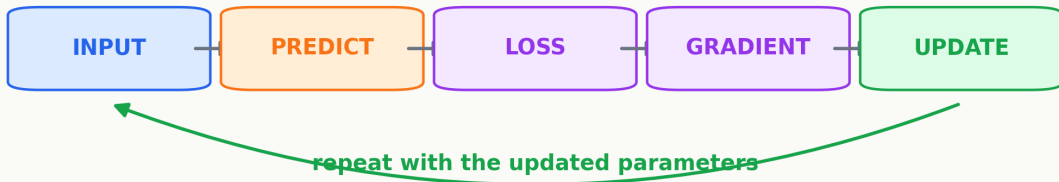
- ▶ Look before using a formula.
- ▶ Name the mistakes you see.
- ▶ Decide what one number should summarize them.

[Open the Google Loss Tutorial](#)

Which line makes the smaller mistakes?



Training repeats the same five ideas



Training is not magic: repeat the loop with the updated parameters.

A data matrix is a table the model can use

- ▶ each row is one house;
- ▶ each input column is one feature;
- ▶ the price is the target.

With 16 houses and 2 features:

X has shape (16, 2).

example	area	distance	price target
house 1	1.74	0.30	2.04
house 2	1.35	1.60	1.33
house 3	1.42	0.30	1.68
house 4	1.68	1.00	1.78

rows = houses

columns = measured values

A linear function makes a prediction

$$\hat{y} = wx + b$$

- ▶ w , the weight, controls the tilt;
- ▶ b , the bias, moves the line up or down;
- ▶ w and b are adjustable parameters.

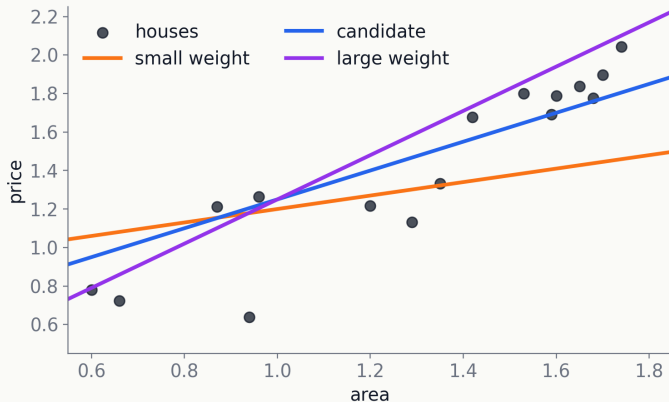


Parameters choose one line from many

Change one number at a time:

- ▶ change w : watch the tilt;
- ▶ change b : watch the whole line shift;
- ▶ goal: reduce the mistake score.

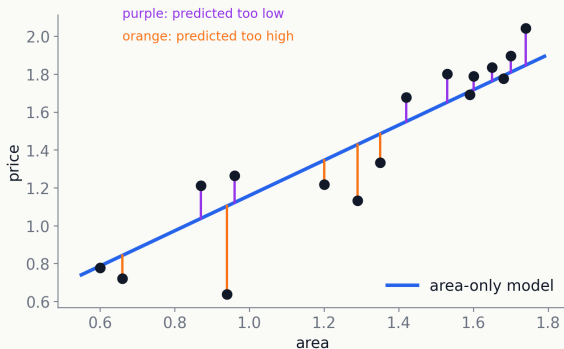
Open the Parameters
Exercise



A residual shows one prediction mistake

$$r_i = \hat{y}_i - y_i$$

- ▶ $r_i > 0$: prediction is too high;
- ▶ $r_i < 0$: prediction is too low;
- ▶ its size is the vertical miss.

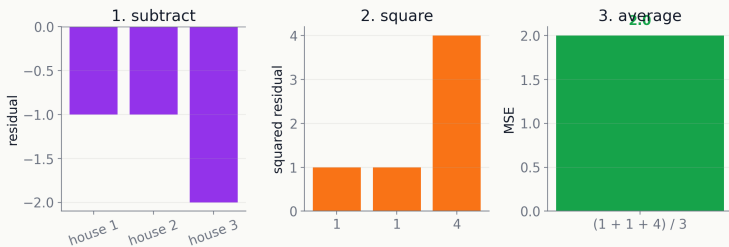


MSE turns many mistakes into one score

For residuals $-1, -1, -2$:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n r_i^2$$

Squaring prevents cancellation.



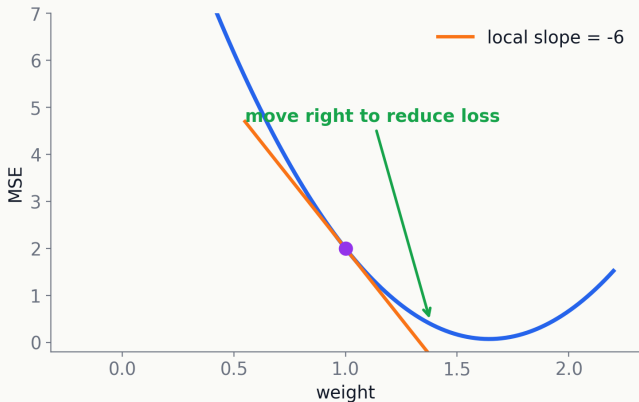
The gradient says which way lowers loss

For one feature:

$$\frac{\partial \text{MSE}}{\partial w} = \text{mean}(2r_i x_i).$$

Open the Gradient Visualizer

- ▶ positive slope: move the parameter left;
- ▶ negative slope: move it right;
- ▶ recalculate after every step.



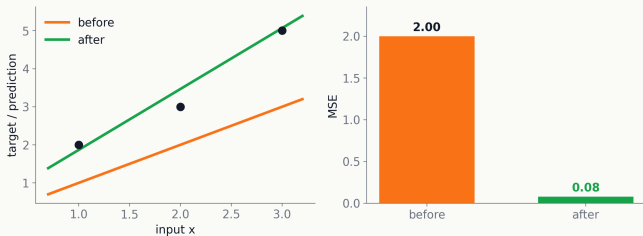
One update can sharply lower the loss

Start with $w = 1$, $b = 0$, and $\eta = 0.1$:

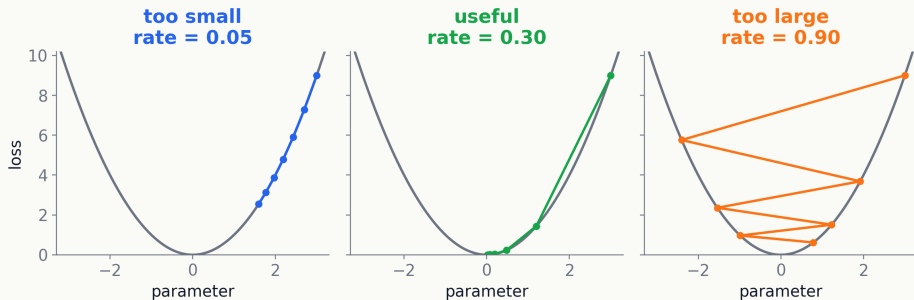
$$g_w = -6, \quad g_b = -\frac{8}{3}.$$

$$w' = 1 - 0.1(-6) = 1.6,$$
$$b' = 0 - 0.1(-8/3) \approx 0.27.$$

MSE: 2.00 $\rightarrow$ 0.08.



Learning rate controls the size of each step



A useful rate moves steadily downhill without jumping across the valley.

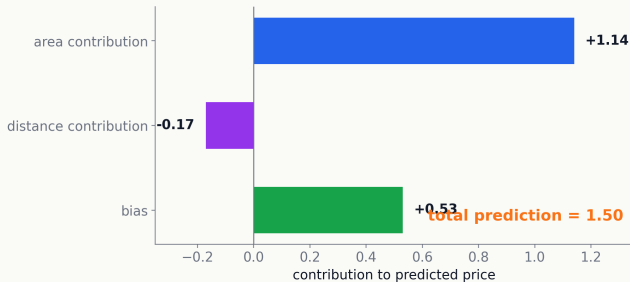
A second feature adds another contribution

$$\hat{y} = w_a x_a + w_d x_d + b.$$

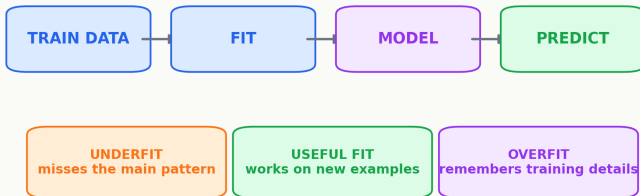
Or, using the dot product from Week 1,

$$\hat{y} = x^\top w + b.$$

Same training loop.

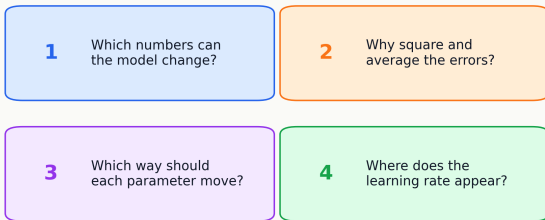


Fitting examples does not guarantee a useful rule



fit learns parameters; predict uses them on inputs.

Explain the update, explain the learning



Return to the [Google Loss Tutorial](#) and explain the better line using **prediction**, **residual**, and **loss**.